

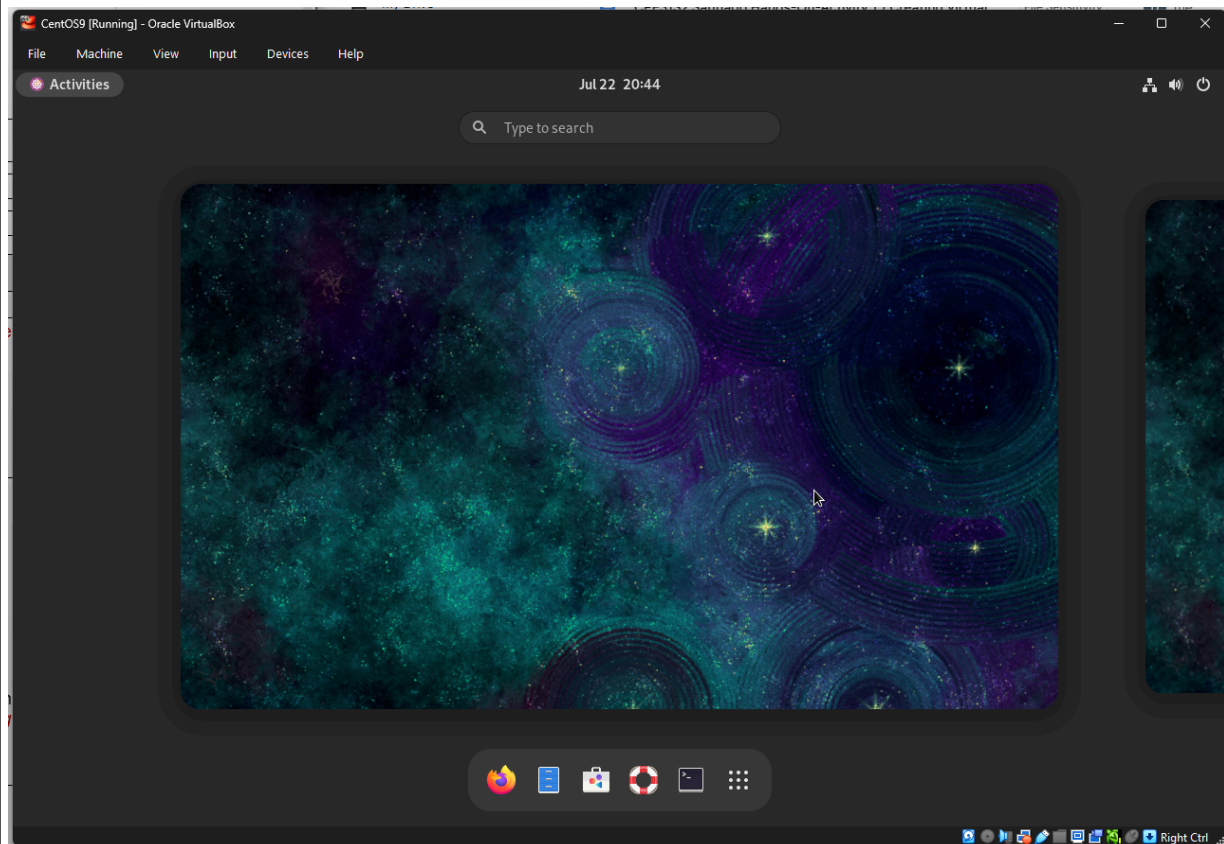
COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 3.1	Hands-on Activity 3.1 - Install SSH server on CentOS or RHEL 8
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1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



1. Install the ssh server package *openssh* by using the *dnf* command:
\$ dnf install openssh-server
2. Start the *sshd* daemon and set to start after reboot:
\$ systemctl start sshd
\$ systemctl enable sshd
3. Confirm that the sshd daemon is up and running:
\$ systemctl status sshd
4. Open the SSH port 22 to allow incoming traffic:
\$ firewall-cmd --zone=public --permanent --add-service=ssh
\$ firewall-cmd --reload
5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
\$ systemctl reload sshd

```
[santiago@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
   Active: active (running) since Wed 2026-07-22 20:43:39 EDT; 3min 22s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 932 (sshd)
      Tasks: 1 (limit: 10502)
     Memory: 1.7M (peak: 1.9M)
        CPU: 8ms
     CGroup: /system.slice/sshd.service
            └─932 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 22 20:43:38 localhost.localdomain systemd[1]: Starting OpenSSH server daemon>
Jul 22 20:43:39 localhost.localdomain sshd[932]: Server listening on 0.0.0.0 po>
Jul 22 20:43:39 localhost.localdomain sshd[932]: Server listening on :: port 22.
Jul 22 20:43:39 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
[santiago@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[santiago@localhost ~]$ firewall-cmd --reload
success
```

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.
3. On CentOS, verify that you have the *authorized_keys*.

```
santiago@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa santiago@centos
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/santiago/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
santiago@centos's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/santiago/.ssh/id_rsa 'santiago@centos'"
and check to make sure that only the key(s) you wanted were added.

santiago@workstation:~$ ssh santiago@centos
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Jul 22 22:08:28 2026 from 192.168.56.101
[santiago@centos ~]$
```

```
[santiago@localhost ~]$ cat ~/.ssh/authorized_keys
cat: ~/.ssh/authorized_keys: No such file or directory
[santiago@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQADmJFJRv+RJYpZVABjuMz1YUnf/mZMLKzTqRkZIJoa4zLnsL0YcBGtIoFZSU
Xw82ITyWQ0/+2SHsKe8tr9tbyC2JkD2yDbi/y2KxHR1fFgcpfmbTW8Jhynz1iCKsoAgZsz2RA8szzdBillyvehHLZYK5+a00I
sBqCMNzj1TgiTS8vBY2AwcjrYhTo0CA7z0jBqqGNuoAZU2ePTXH9dJBflbnQEYmLCxAeJse/KQGx4MHn9x9lDnW4czYGX/HKD
A9A+Wbg8KHuPLBqGuDshWCfD6aFPLGqtXXmf/DiBL6l7+Zhm5V08LF2Im157LjURVMxDmJqfaJJtpUsXqAc32g0shytLvop0LU
Pj80VOPd2ayBSvqje2VD69WLSa5GtCbLYK5k1eoNYMDVOJAHQ+wldJoAUuEk/Ro4ew7LZo4iml7ZHlOGiQ5fRSNcPBFmLE06kn
kZ3QasuUsq2TfP8JhSSAvxyl3fMjYx4HcEwzkD2zQTrxALKlQznmwXj18vgj5yRZkHbJRLWyh9HcUaf8tziZrINM/4U4wbd8s
HSib8fj96W3Hh8peCfdQa4dKWTexn+0wMoiIxML28MlfZx+EnkBWZ9gb8iaC+haBtybg/flhmk0zdu0PW2JL4LuxPkaqoZutcX
f/mmDVffCwvZhFwKN441Yl9dFA/b3qP7K1ZQ== santiago@workstation
[santiago@localhost ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
Last login: Wed Jul 22 22:08:28 2026 from 192.168.56.101
[santiago@centos ~]$ rm hello.txt
[santiago@centos ~]$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
[santiago@centos ~]$ nano hello.txt
[santiago@centos ~]$
logout
Connection to centos closed.
santiago@workstation:~$
```

```
[santiago@localhost ~]$ cd
[santiago@localhost ~]$ ls
Desktop Documents Downloads hello.txt Music Pictures Public Templates Videos
[santiago@localhost ~]$ cat hello.txt
cat: hello.txt: No such file or directory
[santiago@localhost ~]$ cat hello.txt
hello
[santiago@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

The first thing to consider when choosing between a Debian or Red Hat distro are the user needs. For personal use like daily drivers, a Debian distro like Ubuntu would be better due to its user-friendliness, intuitive UI, stability, and compatibility with many software applications and programs. Meanwhile, for commercial oriented needs like enterprise environments, Red Hat distros like RHEL or CentOS would be better due to its compatibility, security, stability, flexibility, and technical support. For a user with simple needs or a desktop environment, Ubuntu would feel the most familiar and comes with all basic features; the user can also freely download anything else that they might need. Meanwhile, for a user with needs for machines like servers or cloud environments, CentOS would be better due to better stability.

2. What are the main differences between Debian and Red Hat Linux distributions?

Both distributions are free and open-source. However, since Red Hat is a subsidiary of IBM, its flagship OS, RHEL, needs to be purchased for technical support and ownership. CentOS serves as RHEL's user feedback source and is the free version of RHEL that is available for all users. Meanwhile, Debian distros, specifically Ubuntu, is free all around including its technical support.

From a technical standpoint, one of the main differences is the download package manager. Ubuntu uses .deb packages and the apt or dpkg package manager. Meanwhile, CentOS uses .rpm packages and the yum/dnf package manager. They also differ in release cycles; Ubuntu offers Long Term Support (LTS) with each version getting released every two years with 5 years of security and maintenance. On the other hand, CentOS has been discontinued and has recently moved on from point updates to the CentOS Stream which follows the RHEL distribution release cycle, with each release being roughly 3 years apart with 5 years of active support and maintenance.

In terms of software compatibility, Debian distributions have more free and open-source programs available in its repository while Red Hat has less, albeit with certification and thorough testing. For flexibility with user needs, Debian provides more options but Red Hat is more secured, certified, and stable.

References:

Which OS is better? Red Hat (RHEL) vs. Debian. (2024, December 4). Ionos Digital Guide.

<https://www.ionos.com/digitalguide/server/know-how/red-hat-vs-debian/>

3. CONCLUSION

In conclusion, this activity introduces the Red Hat version of Linux, CentOS. CentOS is a more stable and tailor-made distribution for enterprise environments and server management. Since this course involves automating servers, CentOS is a better distribution for this task. Basic setup included the VM initialization, enabling SSH, and authenticating an SSH connection between the other VMs through copying the rsa keys. Overall, this activity was straightforward and similar with the setups from the Ubuntu distribution; they only vary with the download package manager and the distro version.

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Purpose of Use: _____

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